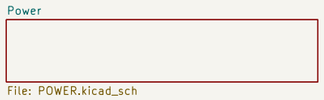
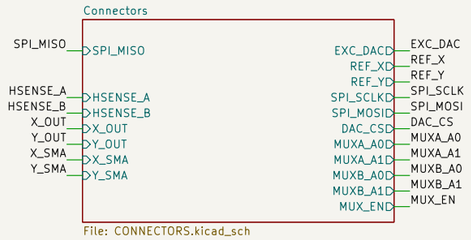
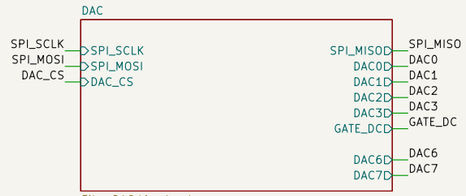
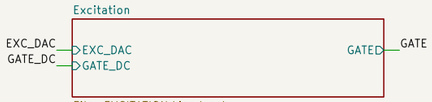
Root sheet: sheet symbols and inter-sheet wiring only (labels of the same name connect). Rails are global power symbols. Pages follow the sheets left to right, top to bottom.

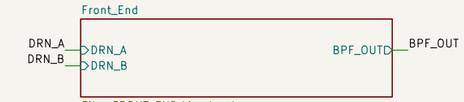














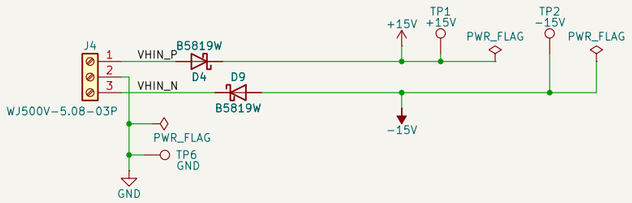
LIA-AFE-Eval

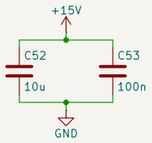
Sheet: /
File: LIA-AFE-Eval.kicad_sch

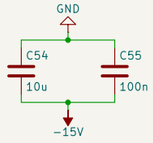
Title: LIA-AFE-Eval: analog front-end evaluation board

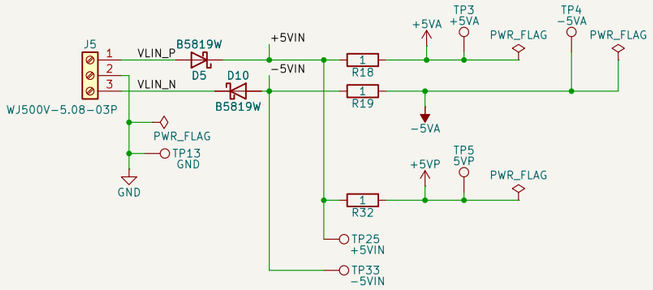
Size: A3	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.8		Id: 1/8

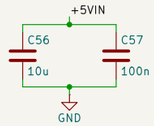
Input: dual lab supply, reverse-polarity protection

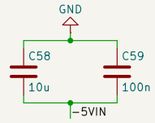








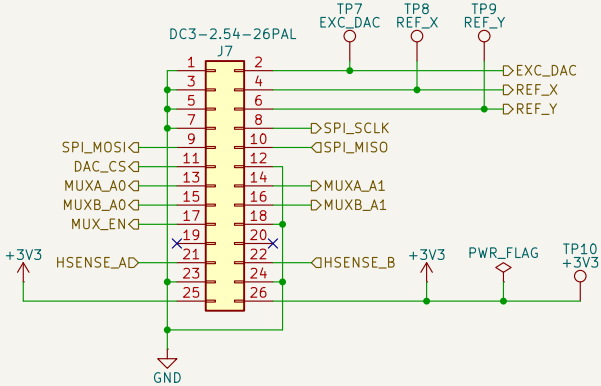


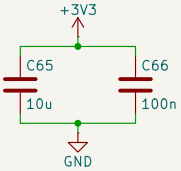


Bench supply note: J5 is set slightly above 5 V so that +5VA (TP3) reads 5.0 V to 5.2 V after the B5819W drop (about 0.3 V to 0.45 V) and R18. Below 5.0 V the DAC80508 internal 2.5 V reference exceeds VDD/2 and trips the reference alarm (all DAC outputs 0 V). Do not let +5VA exceed 5.5 V (DAC80508, TMUX1109).
Power-up order: J4 (+/−15 V, TMUX6136) on first, then J5 (+/−5 V). Power-down in the reverse order.

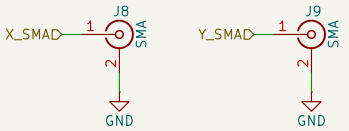
LIA-AFE-Eval		
Sheet: /Power/ File: POWER.kicad_sch		
Title: Power: bench input, protection, analog rails		
Size: A3	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.8		Id: 2/8

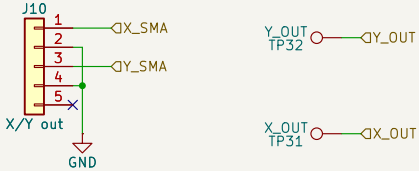
NUCLEO control header: keyed 2 × 13 IDC box header





X / Y outputs: SMA for the scope and a 1 × 5 h





Mounting holes M3, four corners



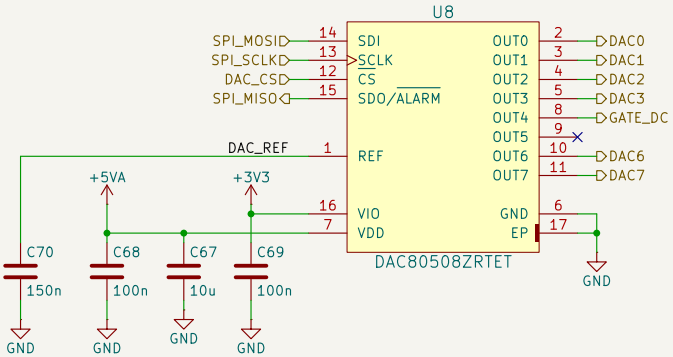
LIA-AFE-Eval

Sheet: /Connectors/
File: CONNECTORS.kicad_sch

Title: Connectors: NUCLEO control header, X/Y outputs, test loops

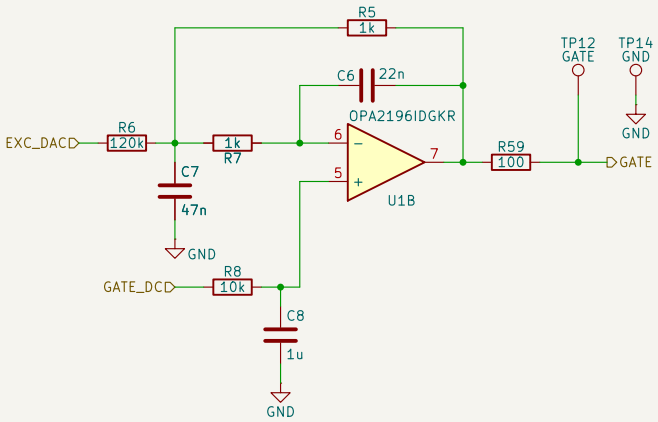
Size: A5	Date:	Rev: 0.1
KiCad E.D.A. 10.0.8		Id: 3/8

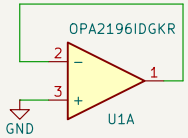
DAC80508 (U8): AVDD = +5VA, VIO = NUCLEO +3V3, internal 2.5 V reference. Channels as LIA-Readout

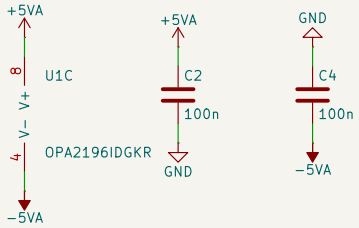


LIA-AFE-Eval		
Sheet: /DAC/ File: DAC.kicad_sch		
Title: Bias DAC: DAC80508 (AN-6)		
Size: A5	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.8		Id: 4/8

Gate drive: U1B → GATE (all pixels, E−11). Gain −1/120 on EXC_DAC, ×1.0083 on GATE_DC, f0 = 4.95 kHz







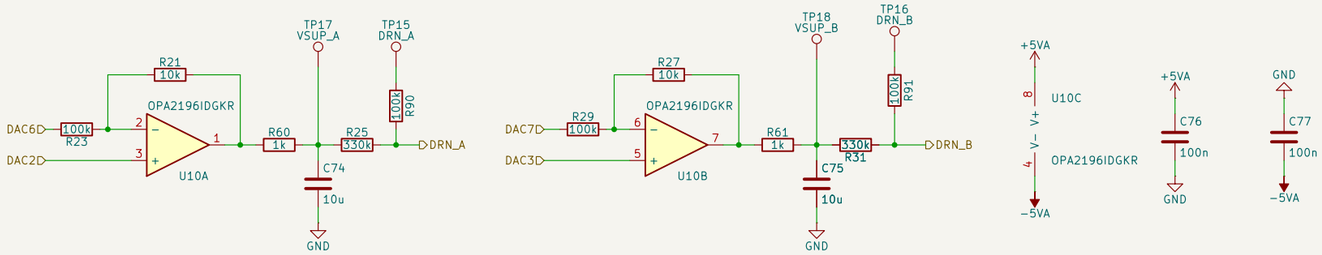
LIA−AFE−Eval

Sheet: /Excitation/
File: EXCITATION.kicad_sch

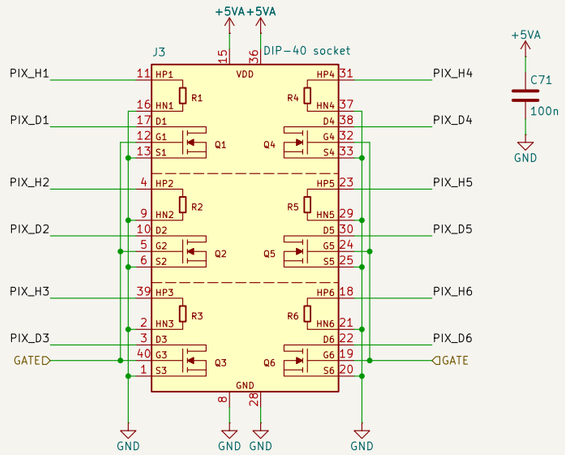
Title: Excitation: split−gate MFB summing stages (AN−3)

Size: A5	Date:	Rev: 0.1
KiCad E.D.A. 10.0.8		Id: 5/8

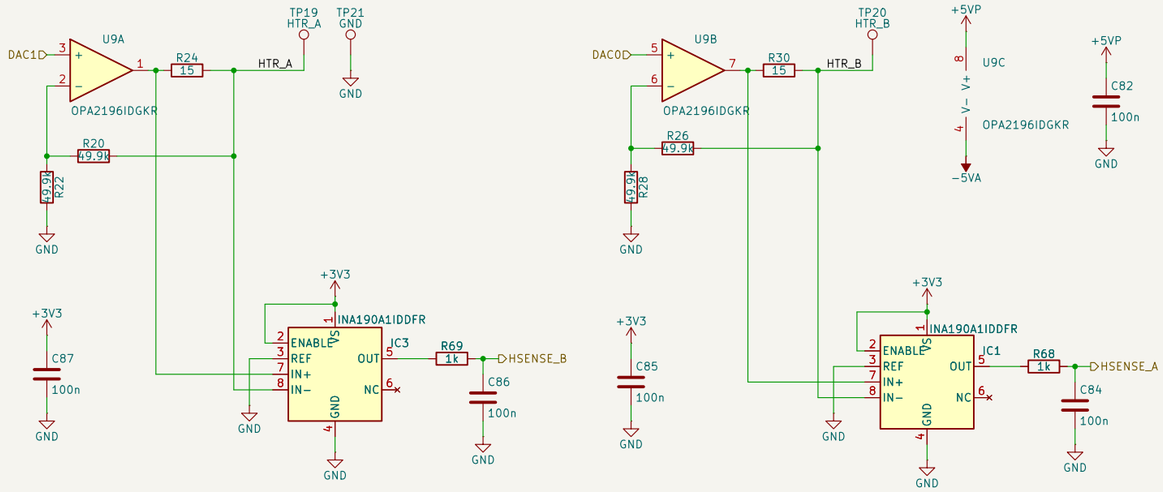
Drain supply



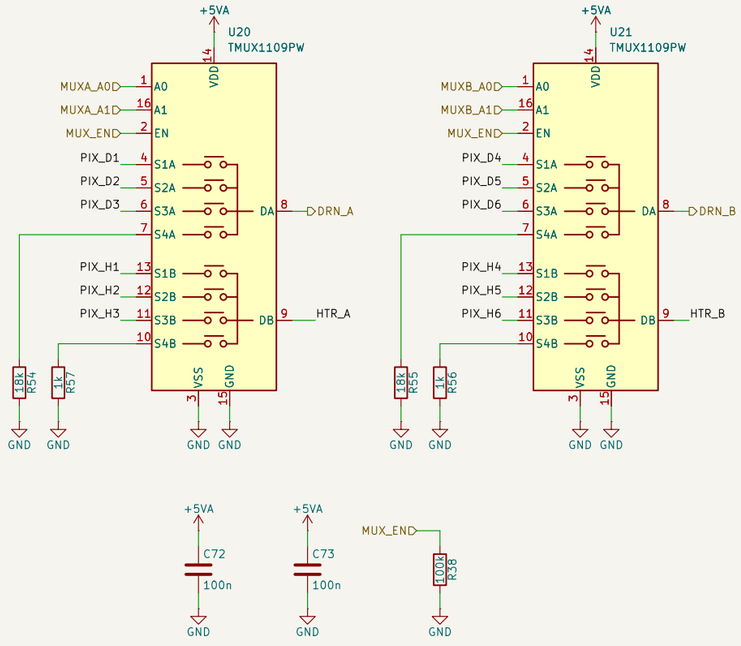
GMOS socket



Heater drivers (gain 2, Kelvin feedback after the shunt), current sense INA190A1



Selection MUX

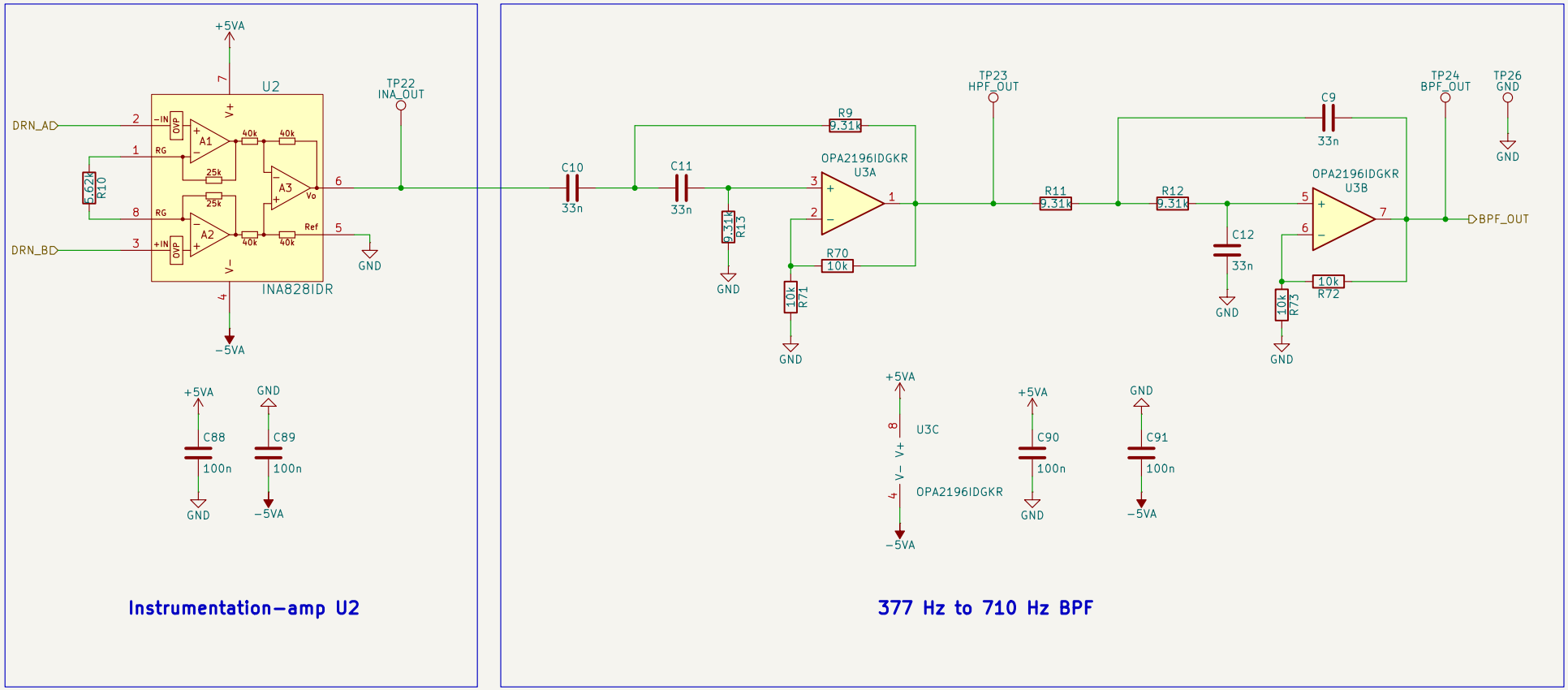


LIA−AFE−Eval

Sheet: /Sensor/
File: SENSOR.kicad_sch

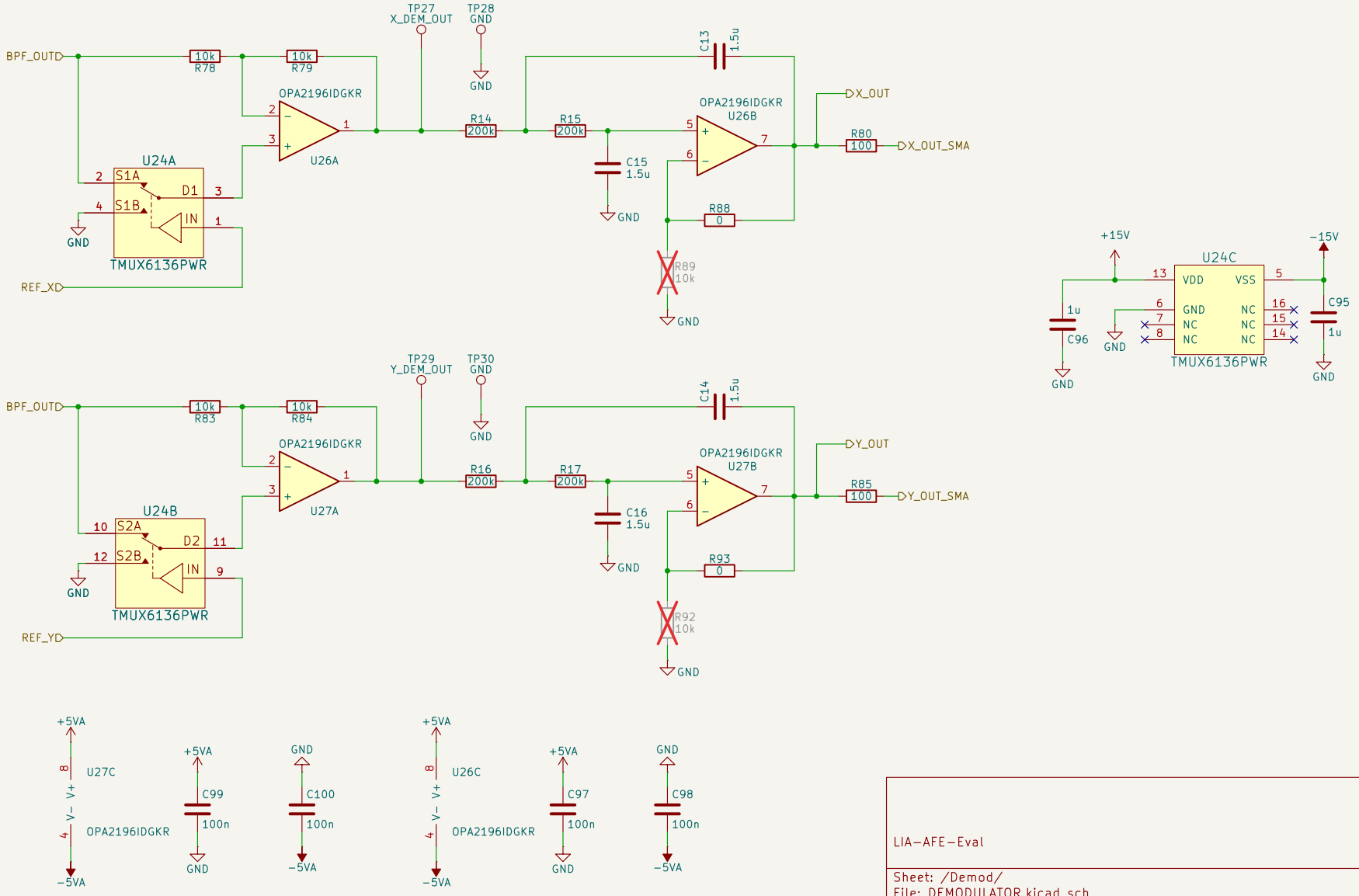
Title: Sensor: GMOS socket, pixel selection, drain bias, heaters

Size: A3	Date:	Rev: 0.1
KiCad E.D.A. 10.0.8		Id: 6/8



LIA−AFE−Eval	
Sheet: /Front_End/ File: FRONT_END.kicad_sch	
Title: Front end FE−B: INA828 G = 9.90, Sallen−Key band−pass 518 Hz	
Size: A4	Date:
KiCad E.D.A. 10.0.8	**Rev: 0.1** Id: 7/8

Switch demodulator: SEL = 1 → S1A = BPF_OUT (gain +1), SEL = 0 → S1B = GND (gain −1). Second switch unused



LIA−AFE−Eval		
Sheet: /Demod/ File: DEMODULATOR.kicad_sch		
Title: Demodulator (Demod_X, Demod_Y): TMUX6136 + OPA2196, gain ±1, po		
Size: A4	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.8		Id: 8/8